

MH1812 – Additional Exercises for Continuous Assessment 2

NTU, AY15/16 S2

Q1: There are 18 mathematics majors and 325 computer science majors at a college.

1. In how many ways can two representatives be picked so that one is a mathematics major and the other is a computer science major?

Solution: $18 \cdot 325 = 5850$ ☐

2. In how many ways can one representative be picked who is either a mathematics major or a computer science major?

Solution: $18 + 325 = 343$ ☐

Q2: Let function $f : X \rightarrow Y$, with $|X| = m$ and $|Y| = n$.

1. How many functions f are possible ?

Solution: n^m ☐

2. How many one-to-one functions f are possible, given $m \leq n$?

Solution: $P(n, m) = n!/(n - m)!$ ☐

3. How many one-to-one correspondence f are possible, given $m = n$?

Solution: $P(n, n) = n!/(n - n)! = n!$ ☐

Q3: How many positive integers less than 1000

1. are divisible by 7?

Solution: $\lfloor \frac{999}{7} \rfloor = 142$ ☐

2. are divisible by 7 but not by 11?

Solution: $\lfloor \frac{999}{7} \rfloor - \lfloor \frac{999}{7 \cdot 11} \rfloor = 142 - 12 = 130$ ☐

3. are divisible by both 7 and 11?

Solution: $\lfloor \frac{999}{7 \cdot 11} \rfloor = 12$ ☐

4. are divisible by either 7 or 11?

Solution: $\lfloor \frac{999}{7} \rfloor + \lfloor \frac{999}{11} \rfloor - \lfloor \frac{999}{7 \cdot 11} \rfloor = 142 + 90 - 12 = 220$ ☐

5. are divisible by exactly one of 7 and 11?

Solution: $\lfloor \frac{999}{7} \rfloor + \lfloor \frac{999}{11} \rfloor - 2 \cdot \lfloor \frac{999}{7 \cdot 11} \rfloor = 142 + 90 - 2 \cdot 12 = 208$ ☐

6. are divisible by neither 7 nor 11?

Solution: $999 - 220 = 779$ □

7. have distinct digits?

Solution: divide into 3 cases according to the number of digits the number has:
 $P(9, 1) + P(9, 1) \cdot P(9, 1) + P(9, 1) \cdot P(9, 2) = 9 + 81 + 648 = 738$ □

8. have distinct digits and are even?

Solution: based on above, divide each of the 3 cases into 2 sub-cases according to even/odd of the first (most significant) digit:

- 1 digit positive even number: 4
- 2 digits positive even number with the first digit odd: $5 \times 5 = 25$
- 2 digits positive even number with the first digit even: $4 \times 4 = 16$
- 3 digits positive even number with the first digit odd: $5 \times 5 \times (10 - 2) = 200$
- 3 digits positive even number with the first digit even: $4 \times 4 \times (10 - 2) = 128$

In total: $4 + 25 + 16 + 200 + 128 = 373$. □

Q4: Solve these recurrence relations together with the initial conditions given.

1. $a_n = a_{n-1} + 6a_{n-2}$ for $n \geq 2, a_0 = 3, a_1 = 6$

Solution: characteristic equation is $x^2 = x + 6$, solving it gives two solutions $x = 3$ and $x = -2$, hence $a_n = u \cdot 3^n + v \cdot (-2)^n$, substituting $a_0 = 3$ and $a_1 = 6$ in, we get $3 = u \cdot 3^0 + v \cdot (-2)^0 = u + v$ and $6 = u \cdot 3^1 + v \cdot (-2)^1 = 3u - 2v$, solving it gives $u = \frac{12}{5}$ and $v = \frac{3}{5}$, hence $a_n = \frac{12}{5} \cdot 3^n + \frac{3}{5} \cdot (-2)^n$. □

2. $a_n = 7a_{n-1} - 10a_{n-2}$ for $n \geq 2, a_0 = 2, a_1 = 1$

Solution: characteristic equation is $x^2 = 7x - 10$, solving it gives two solutions $x = 2$ and $x = 5$, hence $a_n = u \cdot 2^n + v \cdot 5^n$, substituting $a_0 = 2$ and $a_1 = 1$ in, we get $2 = u \cdot 2^0 + v \cdot 5^0 = u + v$ and $1 = u \cdot 2^1 + v \cdot 5^1 = 2u + 5v$, solving it gives $u = 3$ and $v = -1$, hence $a_n = 3 \cdot 2^n - 5^n$. □

3. $a_n = 6a_{n-1} - 8a_{n-2}$ for $n \geq 2, a_0 = 4, a_1 = 10$

Solution: characteristic equation is $x^2 = 6x - 8$, solving it gives two solutions $x = 2$ and $x = 4$, hence $a_n = u \cdot 2^n + v \cdot 4^n$, substituting $a_0 = 4$ and $a_1 = 10$ in, we get $4 = u \cdot 2^0 + v \cdot 4^0 = u + v$ and $10 = u \cdot 2^1 + v \cdot 4^1 = 2u + 4v$, solving it gives $u = 3$ and $v = 1$, hence $a_n = 3 \cdot 2^n + 4^n$. □

4. $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2, a_0 = 0, a_1 = 1$

Solution: characteristic equation is $x^2 = 4x - 4$, solving it gives one $x = 2$, hence $a_n = u \cdot 2^n + v \cdot n \cdot 2^n$, substituting $a_0 = 0$ and $a_1 = 1$ in, we get $0 = u \cdot 2^0 + v \cdot 0 \cdot 2^0 = u$ and $1 = u \cdot 2^1 + v \cdot 1 \cdot 2^1 = 2u + 2v$, solving it gives $u = 0$ and $v = \frac{1}{2}$, hence $a_n = n \cdot 2^{n-1}$. □

Q5: A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.

1. How many balls must she select to be sure of having at least three balls of the same color?

Solution: x balls and 2 colors with $\lceil \frac{x}{2} \rceil \geq 3$, hence $x \geq 5$, at least 5 must she select. $\square$

2. How many balls must she select to be sure of having at least three blue balls?

Solution: 13, worst case is she selects 10 red balls, followed by 3 blue balls. $\square$

Q6: Show that whenever 25 girls and 25 boys are seated around a circular table there is always a person both of whose neighbors are boys.

Solution: We assume that there is a sitting arrangement such that there is no one sitting between two boys. We denote the positions $a_1, a_2, \dots, a_{50}$ so that position a_{50} is next to a_1 . Now we split the youngsters into "odd" and "even" groups: $(a_1, a_3, \dots, a_{49})$ and $(a_2, a_4, \dots, a_{50})$ which we arrange around two smaller tables. Then, by our assumption, no boys are next to each other at either table. So at each table there are at most 12 boys for a total of at most 24 boys. A contradiction. Therefore our assumption was wrong and it is always possible to find someone sitting between two boys. $\square$

Q7: Suppose that 21 girls and 21 boys enter a mathematics competition. Furthermore, suppose that each entrant solves at most six questions, and for every boy-girl pair, there is at least one question that they both solved. Show that there is a question that was solved by at least three girls and at least three boys.

Solution: optional: IMO 2001 Question. For each girl, we know that there is a boy who solved a problem in common with her. Since a girl solves at most six problems, for a girl and her set of six problems there are 21 boys who solved at least one problem in common with the girl such that at least three (or four, $\lceil 21/6 \rceil = 4$) boys solved that common problem. (This is a simple application of the pigeon hole principle where the problems solved by the girl are the (6) holes and the (21) boys are the pigeons, and the guy/pigeon enters a hole/problem which he solved in common with the girl).

The boys have solved a total of at most 21×6 problems. We view this as a set with repeated elements i.e. if a problem is solved by more than one boy it appears as many times as the number of boys who have solved it. Let this set of problems be A. Clearly, A has size 21×6 .

We mark each problem in A which has been solved by at least three boys and a girl. And we mark it the number of times that is the same as the number of girls that has solved it. Since there are a total of 21×11 marks (since there are at least 11 marks for each girl, by the discussion above), either a problem is marked at least thrice, in which case we're done since then it has been solved by at least three boys and three girls. Or each problem has been marked at most twice. In this case it is clear that more than 21×5 problems in A have been marked since $21 \times 5 \times 2 < 21 \times 11$ (there are a total of 21×11 marks. This means that there is a boy such that all six of his problems have been marked. But then by our discussion in the first paragraph we know there

must a problem that this boy has solved which has been solved by at least three girls. Therefore it must be true that there is a problem such that it has been solved by at least three people of each gender. $\square$

Q8: Prove that at a party where there are at least two people, there are two people who know the same number of other people there.

Solution: If there are n people at the party, each one knows somewhere from 0 to $n - 1$ of the others, which is n possibilities. But if one person knows all the others, then all others know at least one person (him/her) (here we assume knowing is reflexive); or if there is one person who knows nobody, then all others will know at most $n - 2$ persons (excluding him/her). Hence the number each person knows is either in the range $[0, n - 2]$ or $[1, n - 1]$. In either case, there are two that match by the pigeonhole principle. $\square$

Q9: How many ordered pairs of integers (a, b) are needed to guarantee that there are two ordered pairs (a_1, b_1) and (a_2, b_2) such that $a_1 \bmod 5 = a_2 \bmod 5$, and $b_1 \bmod 5 = b_2 \bmod 5$?

Solution: There are $5 \times 5 = 25$ pigeonholes, hence 26 pigeons (ordered pairs) are required. $\square$

Q10: Let $(x_i, y_i), i = 1, 2, 3, 4, 5$, be a set of five distinct points with integer coordinates in the xy plane. Show that the midpoint of the line joining at least one pair of these points has integer coordinates.

Solution: we classify the points to $2 \times 2 = 4$ pigeonholes by the rule $(x_i \bmod 2, y_i \bmod 2)$, hence at least 2 will fall in the same pigeonholes, whose midpoint has integer coordinates. $\square$

Q11: Let R be the relation on the set $A = \{1, 2, 3, 4, 5\}$ containing the ordered pairs $(1, 1), (1, 2), (1, 3), (2, 3), (2, 4), (3, 1), (3, 4), (3, 5), (4, 2), (4, 5), (5, 1), (5, 2)$, and $(5, 4)$. Find R^2 and R^3 .

Solution: $R^2 = R \circ R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 4), (2, 5), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$.
 $R^3 = R^2 \circ R = A \times A$. $\square$

Q12: Let R_1 and R_2 be the “congruent modulo 3” and the “congruent modulo 4” relations, respectively, on the set of integers. That is, $R_1 = \{(a, b) | a \equiv b \pmod{3}\}$ and $R_2 = \{(a, b) | a \equiv b \pmod{4}\}$. Find $R_1 \cap R_2$.

Solution: $R_1 \cap R_2 = \{(a, b) | a \equiv b \pmod{12}\}$. $\square$

Q13: Suppose that R and S are reflexive relations on a set A . Prove or disprove each of these statements.

1. $R \cup S$ is reflexive.

Solution: True. $\forall x \in A, (x, x) \in R$ since R is reflexive, and $R \subset R \cup S$, so $\forall x \in A, (x, x) \in R \cup S$, $R \cup S$ is reflexive. $\square$

2. $R \cap S$ is reflexive.

Solution: True. $\forall x \in A, (x, x) \in R$ since R is reflexive, similarly $(x, x) \in S$ as well, so $\forall x \in A, (x, x) \in R \cap S$, $R \cap S$ is reflexive. $\square$

3. $R - S$ is NOT reflexive.

Solution: True. $\forall x \in A, (x, x) \in R$ and $(x, x) \in S$, hence $(x, x) \notin R - S$, $R - S$ is not reflexive. $\square$

4. $S \circ R$ is reflexive.

Solution: True. $\forall x \in A, (x, x) \in R$ and $(x, x) \in S$, hence $(x, x) \in S \circ R$, $S \circ R$ is reflexive. $\square$

Q14: Let R be the relation represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find the matrices that represent R^2 , R^3 , and R^4 .

Solution: $R^2 = R \circ R = \{(1, 3), (2, 1), (2, 2), (3, 2), (3, 3)\}$,
 $R^3 = R^2 \circ R = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$,
 $R^4 = R^3 \circ R = \{(1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$. $\square$

Q15: Find the transitive closures of these relations on the set $A = \{1, 2, 3, 4\}$.

1. $\{(1, 2), (2, 1), (2, 3), (3, 4), (4, 1)\}$

Solution: $R^t = A \times A$. $\square$

2. $\{(2, 1), (2, 3), (3, 1), (3, 4), (4, 1), (4, 3)\}$

Solution: $R^t = \{(2, 1), (2, 3), (2, 4), (3, 1), (3, 3), (3, 4), (4, 1), (4, 3), (4, 4)\}$. $\square$

3. $\{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$

Solution: $R^t = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$ same as R since R itself is already transitive. $\square$

4. $\{(1, 1), (1, 4), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 2)\}$

Solution: $R^t = A \times A$. $\square$

Q16: Determine whether the relations represented by these zero-one matrices are partial orders.

1.

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution: No, R is not transitive because $(2, 1), (1, 3) \in R$ but $(2, 3) \notin R$. □

2.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: Yes. □

3.

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}.$$

Solution: No, R is not transitive since $(1, 3), (3, 4) \in R$, but $(1, 4) \notin R$. □

Q17: Suppose that R_1 and R_2 are equivalence relations on the set S . Determine whether each of these combinations of R_1 and R_2 must be an equivalence relation. a) $R_1 \cup R_2$, and b) $R_1 \cap R_2$.

Solution: $R_1 \cup R_2$ may not be equivalence relation since the “transitive” property may not be satisfied, an example could be $S = \{a, b, c\}$, $R_1 = \{(a, a), (a, b), (b, a), (b, b), (c, c)\}$ and $R_2 = \{(a, a), (b, b), (b, c), (c, b), (c, c)\}$, $(a, b), (b, c) \in R_1 \cup R_2$ but $(a, c) \notin R_1 \cup R_2$. $R_1 \cap R_2$ is equivalence relation. □

Q18: Determine whether the relations represented by these zero/one matrices are equivalence relations.

1.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Solution: NO, since $M(1, 2) \neq M(2, 1)$ hence not symmetric □

2.

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Solution: Yes. Diagonal are all “1”s, hence “reflexive”; Matrix is symmetric with respect to the diagonal, hence “symmetric”, “transitive” can be checked manually. □

3.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Solution: Yes with the same reasoning as above. $\square$

Q19: Let $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$, and define functions $f, g : \mathbb{Z}_5 \rightarrow \mathbb{Z}_5$ as follows: $f(x) = (x + 4)^2 \pmod{5}$, and $g(x) = (x^2 + 3x + 1) \pmod{5}$. Is $f = g$?

Solution: Yes, because $f(x) = (x + 4)^2 = x^2 + 8x + 16 \equiv x^2 + 3x + 1 \pmod{5} = g(x)$. $\square$

Q20: Determine whether the following functions defined on a set of real numbers are one-to-one:

1. $f(x) = \frac{x+1}{x}$, for all real numbers $x \neq 0$

Solution: Let $\frac{x_1+1}{x_1} = \frac{x_2+1}{x_2}$ for non-zero real numbers x_1, x_2 , then $x_2(x_1 + 1) = x_1(x_2 + 1)$, hence $x_1 = x_2$, f is one-to-one. $\square$

2. $f(x) = \frac{x}{x^2+1}$, for all real numbers x

Solution: Let $\frac{x_1}{x_1^2+1} = \frac{x_2}{x_2^2+1}$ for some real numbers x_1, x_2 , then $x_1(x_2^2 + 1) = x_2(x_1^2 + 1) \rightarrow (x_1 - x_2)(1 - x_1x_2) = 0$, hence $x_1 = x_2$ **or** $x_1 = \frac{1}{x_2}$, hence f is not one-to-one. $\square$

Q21: find g^{-1} , f^{-1} , $g \circ f$, $f \circ g$, for $f, g : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = x + 3$, $g(x) = -x^3$.

Solution: $g^{-1}(x) = \sqrt[3]{-x}$, $f^{-1}(x) = x - 3$, $g \circ f(x) = -(x + 3)^3$, $f \circ g(x) = -x^3 + 3$ $\square$

Q22: What is the range of $f : \mathbb{R} \rightarrow \mathbb{R}$, and $f(x) = x^2 + 6x - 8$.

Solution: $f(x) = (x + 3)^2 - 17 \geq -17$, hence range is $[-17, +\infty)$ $\square$